

Math 8 Mid-Q4 Test

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Name: _____ Class: _____ Date: _____

Directions: Answer all 20 questions in the space provided.

1. $V = 15 \text{ V}$, $R = 3 \Omega$. I ?

Answer: _____

2. $I = 6 \text{ A}$, $R = 4 \Omega$. V ?

Answer: _____

3. $V = 24 \text{ V}$, $I = 6 \text{ A}$. R ?

Answer: _____

4. $P = VI$. $V = 5$, $I = 4$. P ?

Answer: _____

5. $v = f\lambda$. $f = 6 \text{ Hz}$, $\lambda = 4 \text{ m}$. v ?

Answer: _____

6. $v = 200 \text{ m/s}$, $f = 20 \text{ Hz}$. λ ?

Answer: _____

7. If λ doubles (constant v), f ?

Answer: _____

8. Higher frequency light $\rightarrow$ wavelength ?

Answer: _____

9. r doubles $\rightarrow T \times$?

Answer: _____

10. Earth $T=1 \text{ yr}$. Planet $r=25 \text{ AU}$. T ?

Answer: _____

11. Planet $r = 16 \text{ AU}$. T ?

Answer: _____

12. Ohm's Law?

Answer: _____

13. Wave equation?

Answer: _____

14. Higher $R \rightarrow I$ is ?

Answer: _____

15. Kepler 3rd Law relates T to ?

Answer: _____

16. 100 W bulb for 20 hrs. Energy?

Answer: _____

17. 400 nm light in m?

Answer: _____

18. $v = 340 \text{ m/s}$, $f = 340 \text{ Hz}$. λ ?

Answer: _____

19. Larger orbit = period is ?

Answer: _____

20. Kepler's 3rd Law: $T^2 = ?$

Answer: _____